



American International University-Bangladesh (AIUB)
Faculty of Science and Technology

CyberRiskXAI: An Explainable Machine Learning Framework for Cybersecurity Risk Prediction and Risk Aware Recommendations

MD Arman Islam (22-49141-3)

MD. Rizvi Karim Ohy (22-49544-3)

Naimur Rahman Monyem (22-49547-3)

Sazia Afrin Dima (22-48402-3)

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Declaration

This thesis is composed of our original work, and contains no material previously published or written by another person except where due reference has been made in the text. We have clearly stated the contribution of others to our thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financial and any other original research work used or reported in our thesis. The content of our thesis is the result of work we have carried out since the commencement of the Thesis.

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MD Arman Islam

22-49141-3

Department of Computer science &
Engineering

MD. Rizvi Karim Ohy

22-49544-3

Department of Computer science &
Engineering

Naimur Rahman Monyem

22-49547-3

Department of Computer science &
Engineering

Sazia Afrin Dima

22-49402-3

Department of Computer science &
Engineering

Approval

The thesis titled “CyberRiskXAI: An Explainable Machine Learning Framework for Cybersecurity Risk Prediction and Risk Aware Recommendations .” has been submitted to the following respected members of the board of examiners of the department of computer science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science on **August 6, 2026** and has been accepted as satisfactory.

Syed Nafiul Shefat

Lecturer & Supervisor

Department of Computer Science

American International University-Bangladesh

Sharfuddin Mahmood

Assistant professor & External

Department of Computer Science

American International University-Bangladesh

Prof. Dr. Muhammad Firoz Mridha

Professor & Head (UG)

Department of Computer Science

American International University-Bangladesh

Prof. Dr. Khandaker Tabin Hasan

Professor & Associate Dean

Faculty of Science and Technology

American International University-Bangladesh

Prof. Dr. Dip Nandi

Professor & Dean

Faculty of Science and Technology

American International University-Bangladesh

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List of Abbreviations

2FA	Two-Factor Authentication
AI	Artificial Intelligence
API	Application Programming Interface
BDT	Balanced Decision Tree
CSF	Cybersecurity Framework
CSV	Comma-Separated Values
DL	Deep Learning
GNB	Gaussian Naive Bayes
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
ISMS	Information Security Management System
LIME	Local Interpretable Model-Agnostic Explanations
LR	Logistic Regression
ML	Machine Learning
NIST	National Institute of Standards and Technology
RBAC	Role-Based Access Control
RF	Random Forest
SE	Software Engineering
SHAP	SHapley Additive exPlanations
TPU	Tensor Processing Unit
UI	User Interface
XAI	Explainable Artificial Intelligence

Abstract

Cybersecurity risks are often caused not only by technical weaknesses but also by unsafe user behaviour. This study developed CyberRiskXAI, an explainable machine learning framework for predicting human-related cybersecurity risk. The system classifies users into Low Risk, Medium Risk, and High Risk groups, explains the reasons behind each result, and provides practical security recommendations. Data were collected through a structured survey covering password reuse, two-factor authentication, phishing awareness, unknown links, password sharing, public Wi-Fi, antivirus use, privacy habits, secure website checking, password management, and software installation. The data were cleaned, encoded, and converted into numerical values. Safe behaviour received higher scores, while unsafe behaviour received lower scores. A Cyber Safety Score and a Cyber Risk Score were then created. The dataset was divided into 80% training data and 20% testing data using stratified sampling. Four supervised machine learning models were compared: Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression, and Random Forest. Logistic Regression achieved the best performance, with 97.47% accuracy, 99.02% precision, 90.48% recall, and a 93.95% F1-score. SHAP and LIME were used to explain overall and individual predictions. The selected model was integrated into a Streamlit web application for both individual and enterprise-level assessment. The results show that survey-based machine learning can identify human cybersecurity risk with strong overall performance. The framework also improves transparency by linking predictions with clear explanations and recommendations.

Keywords

Cybersecurity, Cybersecurity Awareness, Cybersecurity Risk Prediction, Machine Learning, Explainable Artificial Intelligence, Logistic Regression, SHAP, LIME, Human Risk Assessment

Chapter 1

Introduction

1.1 Introduction

Digital technology has come a long way and has transformed the way that people and organizations work in today's digital world. Technologies such as cloud computing, mobile devices, artificial intelligence, and online platforms have improved efficiency and communication across various sectors. Industries such as education, banking, healthcare & government, corporate. But there are also new cybersecurity challenges, as well, with the growing reliance on digital systems. There are various kinds of cyber threats such as phishing, malware, ransomware, identity theft, and unauthorized access. They are still impacting individuals and organizations, resulting in financial losses, data breaches and disruption of operations (Alharbi & Tassaddiq, 2021).

The majority of traditional cyber security awareness evaluations are based on questionnaires and statistical methods to gauge user knowledge and security behavior. These methods have been used in numerous investigations. These studies include things like password security, phishing awareness, Internet safety and social media practices (Alqahtani, 2022). These approaches typically yield descriptive results, however, and do not necessarily offer predictive insights. They can provide the current awareness level of users, but can't foresee future cybersecurity risks accurately. Furthermore, there are multiple cybersecurity awareness campaigns that offer generic advice without taking user differences into account. Employees in a banking organization, corporate office or government environment, for instance, might be exposed to different cybersecurity risks based on their roles and online activities. Therefore, a more intelligent approach is needed. The method will be capable of analyzing the data related to cyber security, predicting the threats, and offering customized suggestions. Machine learning is a promising solution. It can uncover hidden patterns in data and assist with cybersecurity prediction tasks, such as threat detection, risk assessment, and user behavior analysis (Abbas et al., 2025).

Many models are not comprehensible to the user as they are black box systems. A prediction with no explanation and trust may be less and use may be limited. Explainable Artificial Intelligence (XAI) provides a solution by improving the transparency of machine learning models. It will be able to demonstrate how factors can impact outcomes of

prediction. Explain overall model behaviour and individual predictions using techniques like SHAP and LIME. This can assist users in comprehending the rationale behind a specific cyber security risk level. This research aims to introduce an Explainable Machine Learning Framework for Cybersecurity Risk Prediction and Risk-Aware Recommendations. The suggested framework is meant to serve various sectors such as banking, corporate organizations and educational institutions. It will be offering a clear means to forecast cybersecurity dangers, clarify the origins of cyber dangers, and recommend appropriate cybersecurity enhancements. This research aims to enhance cybersecurity decision-making processes and promote safer digital practices through the integration of predictive analysis, explainability, and personalized guidance.

1.2 Problem Statement

As digital technologies become more and more important, individuals and organizations in various sectors have become exposed. Industries such as banking, corporate, educational and government. Even with the evolution of security technologies, there are many incidents taking place because of human-related factors. Poor security habits, lack of awareness and unsafe online behavior. The current cybersecurity assessment methods primarily aim to measure the level of awareness via surveys and statistical analysis. However, they offer little insight into the actual level of risk that an individual may pose to cyber security or what individual factors may be at play(Alharbi & Tassaddiq, 2021; Alqahtani, 2022).

Many of the existing machine learning based prediction approaches in cybersecurity are not transparent. They frequently make predictions, but don't explain how they make decisions. This constraint hinders user and organization understanding of the rationale for identified risks. Hence, the present challenge is the lack of a proper approach to assess the cybersecurity risk in an accurate manner. Also will be able to give comprehensible clues about the factors affecting the risk prediction (Alshaikh & Adamson, 2021).

There is a lack of an integrated and explainable approach for cybersecurity risk prediction. It complicates the process of organizations to determine the vulnerable users and make informed decisions regarding cybersecurity. Existing approaches do not adequately cover the need for transparent risk assessment to get meaningful insight into cybersecurity weaknesses. This research aims at the issue of poor prediction performance and lack of interpretability of existing cyber security risk assessment approaches.

1.3 Problem Background

The growth of digital transformation has changed the way organizations manage their operations and interact with users. Digital information is gathered and processed in vast quantities by business and institutions now through internet platforms. While this change has made things more efficient, it has also made it more complex to manage cybersecurity risks. Cybersecurity protection, traditionally, has been centered on technical solutions. This include network monitoring, access control, encryption, and security software. But from many cybersecurity incidents, it is learned that technical protection is not enough. Digital users may unwittingly introduce vulnerabilities. For this reason, organizations have implemented cybersecurity awareness programs and user training to enhance security behavior and minimize the risks associated with humans. In addition to awareness programs, organizations have begun to gather and analyze cybersecurity data related to users. Previous studies have mainly used surveys and statistical methods. These methods have helped in understanding the cybersecurity behaviour. Meanwhile, machine learning has opened up new opportunities for working with complex cybersecurity data. It also helps to make intelligent decisions in security management.

Cybersecurity behavior analysis and intelligent data-driven techniques are a new research field. There is a growing need for organizations to have methods to help them manage their cybersecurity proactively. This has led to interest in approaches to predictive models and explainable AI. It can help the security professional to comprehend risks and make informed decisions.

1.4 Research Objectives

The primary goal of this study is to build a machine learning-based framework for cybersecurity risk prediction and risk-aware recommendations that is explainable. The proposed framework will focus on analyzing cybersecurity related data and predicting the cybersecurity risk level for the users. It also seeks to explain the factors affecting the prediction results and offer the right suggestions. These must drive an improvement in cyber security practices among various sectors.

The specific objectives to achieve the main goal are as follows:

1. To identify and gather cybersecurity related data in terms of factors related to cybersecurity awareness, knowledge and online security behavior of users in this context.

2. To preprocess and transform collected data with suitable data preparation techniques to enhance the quality and suitability of the collected dataset for machine learning analysis.
3. To build a machine learning based cybersecurity risk prediction model to classify users based on their cybersecurity risk level using Logistic Regression.
4. To assess the performance of the developed prediction model with the help of appropriate evaluation metrics to determine its effectiveness in predicting cybersecurity risks.
5. To use Explainable Artificial Intelligence (XAI) methods such as SHAP and LIME to explain the most important factors influencing the predictions and increase the interpretability of the machine learning model.
6. To provide users with cybersecurity recommendations to help them enhance their security practices, generated based on the results of the prediction and risk factors identified.
7. Develop and deploy a web-based decision support dashboard that communicates in an understandable and accessible way cybersecurity risk predictions, explanations and recommendations.

The objectives focus on overcoming previous cybersecurity assessment approaches. It will incorporate risk prediction, model explanation, and personalized guidance all in one. The goals are within the scope of this research, and suitable for the development of the proposed CyberRiskXAI framework.

1.5 Research Questions

The research questions that drive this research are directly related to the research objectives. The questions aim at exploring the analysis of cybersecurity-related data, the use of machine learning for cybersecurity risk prediction, and the enhancement of the understanding of the prediction results using Explainable Artificial Intelligence techniques. These questions aim to build, validate and enhance the proposed CyberRiskXAI framework for cybersecurity risk prediction and risk-aware recommendations.

Research questions are:

Q1: What are the user data cybersecurity aspects that can be identified to support cybersecurity risk assessment and prediction?

Q2: What are some of the ways the data related to cybersecurity can be preprocessed and transformed to enhance the performance of the machine learning based risk prediction model?

Q3: What is the performance of a Logistic Regression model in predicting the cybersecurity risk levels from users' cybersecurity awareness and behaviour related features?

Q4: What are the possible benefits of using Explainable Artificial Intelligence (XAI) techniques (like SHAP and LIME) to enhance interpretation and understanding of cybersecurity risk prediction results?

Q5: How to use the outcomes of the prediction and the explainability results to create personalized risk-aware cybersecurity recommendations?

Q6: How can a web-based decision support dashboard be designed to present cyber security risk predictions, explanations and recommendations in an understandable manner?

The following research questions are used to provide a guide for developing and testing the proposed framework. All of the questions are working toward the overall research goals.

1.6 Motivations

The impetus for this research is the rising effect of cybersecurity threats on organizations and individuals. Digital technologies are essential to people's everyday lives. Banking, corporate, healthcare and education are some of the sectors that are constantly implementing digital systems. But this digital dependence has also led to a rise in cyber risks. Here human-related factors remain one of the major challenges in maintaining cybersecurity. There are plenty of organizations that offer security awareness training and security training. But existing approaches mainly focus on measuring cybersecurity knowledge and behavior. Should target prediction of potential risks and the specific factors that contribute to those risks.

In addition, machine learning's application in cybersecurity can be used to monitor and anticipate potential threats by examining user behavior. The output from many of the existing prediction models is not well explained, meaning that it is hard for the user and/or organization to understand why the model has assigned a specific risk level. This opacity can have an impact on trust and reduce the effective application of machine learning cyber security systems. Hence, a more comprehensible and practical approach is necessary. Such

an approach will be based on prediction, explanation and helpful suggestions. The motivation for this research is to overcome the gap between cybersecurity risk prediction and human understanding. It can be done through the creation of an explainable framework that can help inform more security decisions. This research seeks to help improve cybersecurity risk management strategies by combining machine learning with Explainable Artificial Intelligence (XAI). It can assist organizations in various industries to make well-informed security choices and enhance their cybersecurity practices.

1.7 Flow of Research

This research has been carried out in a systematic process with 5 stages. Each phase is geared towards the accomplishment of the research goals and is structured to allow for a proper research process. The research starts from Phase 1: Problem Identification and Literature Review. In this stage, there is an effort to get information about the current cyber security problems and to review the existing studies. The aim of this stage is to find out the gaps in the current approaches and determine the research gap. Once the problem of the research is identified, it moves on to Phase 2: Framework Design. In this phase, the proposed framework is formulated in terms of the requirements and the research objectives mentioned in the previous step. The design phase sets up the framework structure. This makes sure that every element of the framework is effective in advancing the research goal. The designed framework is then developed during Phase 3: Implementation. This step is about turning the proposed framework into a working system. The implementation process is the application of the designed approach to the data related to cyber security, and the development of the components needed. The outcome of this phase is a working framework that can be further evaluated. Upon implementation, the research is ongoing in Phase 4: Evaluation and Validation. This stage explores the effectiveness and performance of the created framework. The evaluation process is to determine if the framework can effectively make cybersecurity risk predictions and offer insightful explanations and recommendations. Analysis of the results from this step is used to determine strengths and weaknesses of the proposed approach.

Lastly, the research ends with the step of discussion and conclusion (Phase 5). This phase provides the results of the study, a discussion of the research contributions and possible directions for future improvement. The final stage represents the overall results of the research and shows the contribution of the proposed framework in the risk assessment and management of cybersecurity.

1.8 Significance of the Research

The relevance of this research is due to the increased demand for cyber security risk assessment that is more effective and easily understood. The proposed framework can offer organizations practical benefits, through increasing their understanding of cyber security. A more structured way to identify potential could be helpful for industries like banking, corporate organizations, educational institutions, and other industries that rely on digitalization. The framework can help organizations identify areas that need improvement by offering risk predictions as well as explanations and recommendations. It creates more focused cybersecurity awareness and training initiatives. This research can help to understand the vulnerabilities people have and what factors might make them more vulnerable. Users will not be provided with general security advice, but security recommendations. This can help users develop safe online behavior and raise awareness of cybersecurity risks.

The results of this research add to the body of work in cybersecurity analytics and focus on the use of machine learning and Explainable Artificial Intelligence. It serves as a stepping stone for future researchers who want to create intelligent, transparent and user-centric cybersecurity solutions. The study also highlights the importance of combining prediction capabilities with explainability. In sum, this study has both practical and academic significance, as it can facilitate the understanding of cybersecurity risk and enhance decision-making.

1.9 Research Contribution

This study makes a valuable contribution in the area of cyber security by providing an integrated framework comprising of cybersecurity risk assessment, machine learning based prediction and Explainable Artificial Intelligence (XAI). The traditional cybersecurity awareness studies, which are based on the measurement of users' knowledge and security behavior. This research is an extension of the assessment process. It presents a predictive method that can help in determining the level of cybersecurity risk and give a sense of the factors that can impact that risk.

This research has the following specific contributions:

1. This research introduces a structured framework. It is a link between the collection of

cybersecurity data, data preparation, risk prediction, explainability, and recommendation generation. The framework offers a full process to analyze cybersecurity risks.

2. The approach used in this study is machine learning-based approach to analyze cybersecurity related factors and to predict users' risk levels. This helps complement current cybersecurity assessment methods with the ability to make predictions to identify potential risks.

3. This study combines the SHAP and LIME approaches for explaining predictions from machine learning. These techniques are employed to determine the factors that affect cybersecurity risk predictions. It also gives easy to understand explanations of individual risk assessments, overcoming the opacity of traditional black-box models.

4. This research provides a recommendation approach. Based on the results of the prediction and the risk factors identified, it offers custom-made cyber security guidance. The framework is not a security checklist, but rather a way to offer individual users personalized security suggestions, tailored to their unique cybersecurity vulnerabilities.

5. Web-based implementation of the proposed framework is also presented in this study. The platform offers easy-to-understand risk insights, explanations, and suggestions. These help users and organizations to better understand and manage cybersecurity risks.

The overall impact of this work is to provide fresh knowledge by combining prediction, explanation and recommendation in one. The proposed approach is more transparent and user friendly. It can be customized for various industries, such as banking, corporate, education, and other digital-dependent environments, and is used for cybersecurity decision-making.

1.10 Thesis Organization

This thesis consists of six chapters. One chapter for each of the parts of the research process.

Chapter 1: Introduction contains the research background, problem statement, research objectives and research questions. It also includes motivation, research flow, significance, contributions, and organization of thesis.

The existing studies are discussed in chapter 2: Literature Review. Studies related to cybersecurity risk assessment, machine learning applications and Explainable Artificial Intelligence are discussed here. It can be used to build a research base and determine the research gap.

The design and structure of the proposed system is introduced in chapter 3: Proposed

Model. It is comprised of Explainable Machine Learning Framework for cybersecurity risk prediction and risk-aware recommendations. It also contains the significant elements and connections.

The development of the proposed framework is explained in chapter 4: Implementation and Testing. It covers the system implementation, model development, testing procedures and evaluation of the framework performance.

The standards that have been taken into account during the research are discussed in Chapter 5: Standards, Constraints and Milestones. It shows what restrictions and constraints that influenced the study and planned milestones that were carried out during the research process.

The final chapter (Chapter 6: Conclusion) presents the findings of the research overall. It showcases the key contributions, addresses the limitations, and suggests potential avenues for future enhancements.

1.11 Summary

The background of the study was provided in this chapter. It includes the background and significance of Cyber Security Risk Assessment in various industries. The chapter identified the limitations of existing cybersecurity assessment approaches. It also highlighted their lack of skills in predicting individual cyber threats and giving clear explanations. Based on this problem, the research aims to develop an Explainable Machine Learning Framework for Cybersecurity Risk Prediction and Risk-Aware Recommendations.

The chapter also presented the research objectives and questions that guide the development of the proposed framework. The study aims to analyze the data related to cyber security and to build a risk prediction model. This model aims to make the prediction transparent by the use of Explainable Artificial Intelligence techniques and provide the risk-aware recommendations. Furthermore, the motivation, the research flow, the significance and contribution of the study were discussed to accentuate the practical and academic importance of the study.

Lastly, an outline of the thesis was given to give an overview of the structure and flow of the research. The following chapters will discuss the literature, proposed model development, implementation and testing, research standards and constraints and the overall conclusion of the study. This is a structure that will give the research direction. It

provides for systematic attainment of research objective that will aid in solving the challenges of cybersecurity risk prediction.

Chapter 2

Literature Review

2.1 Introduction

The research background, objectives and scope were introduced in the previous chapter. It also brought to attention the importance of creating an intelligent and transparent attitude in cybersecurity risk management. This chapter will analyze the research problem in detail and review the existing literature. Machine learning models can be used to enhance threat detection and risk prediction. Many of the advanced models can be complex, making it hard for security analysts to interpret, validate and trust their results (Mohale & Obagbuwa, 2025; Prasad et al., 2026). This lack of explainability is particularly important in cybersecurity risk assessment.

The chapter emphasizes three key areas: quantitative cybersecurity risk assessment, application of a machine-learning and deep-learning model for cybersecurity risk prediction, and explainable AI techniques like SHAP and LIME. Past research demonstrates that intelligent models can effectively detect the threats and predict the level of risk. Explainability methods can then be used to provide insight to a user on which factors affected the model's prediction (Almheiri et al., 2025; Salloum & Norozpour, 2025). But the accuracy of the results does not make a model fully useful. Its predictions should also be understandable to security professionals to be used for decision-making. Some problems have also been identified. The issues include data quality issues, lack of explanation reliability, high processing demands, and low performance in real-world scenarios. These make it difficult in providing suitable security recommendations for different risk levels (Mohale & Obagbuwa, 2025). The key ideas, techniques, data and results of previous research are summarized in Section 2.2. The strengths and weaknesses of the existing research and gaps are discussed in Section 2.3. It explains why there is not a clear relationship between risk prediction, explanation of the model and security suggestions. Finally, in Section 2.4 the chapter is summarized and it is explained how the proposed CyberRiskXAI framework can overcome these research gaps.

2.2 Literature Review

Cybersecurity risk assessment can assist organizations in determining threats and estimating potential damages. Expert judgement is often applied in traditional approaches and terms like low, medium, or high risk are employed. These techniques are simple to use and can be subjective. They can also have problems with massive quantities of security-changing data. This can be minimized by the use of quantitative methods which measure the risk by using numbers. For instance, probability and linear programming can aid in calculating the risk and choosing an appropriate control that is cost-effective. But these methods rely on trustworthy threat, vulnerability, and control effectiveness information. This may not be the case in actual organizations. Thus, quantitative risk assessment can help to increase consistency, and the degree of reliability of this approach will depend on the data that are available for the calculation (Sahinoglu, 2024).

Machine learning is an essential tool to analyze the complex data in the field of cybersecurity. Can detect patterns that might not be apparent during manual analysis. It has been used in intrusion detection, malware classification, vulnerability assessment, fraud detection and prediction of cyber risk. One smart risk-assessment approach involved comparing centralized and federated machine-learning models for Internet of Things systems. The prediction results of the models like XGBoost and LightGBM were good. The federated models also worked well, and had the advantage of providing greater privacy, since data stayed on the local devices. It is helpful in distributed environments where sensitive data can't be stored in a single location. But synthetic data were primarily used in the study. Its outcomes can thus vary when used to deploy the models in a real industrial network. In reality, there are missing values, unknown devices, unusual traffic and even attacks you haven't seen in real networks (Abbas et al., 2025).

Machine learning can also be used to enhance cloud security by detecting intrusions, identifying malicious activities, protecting sensitive information, and preventing fraud. But challenges include poor data quality, high processing costs and a lack of adaptation to new threats. Sometimes a risk-prediction model might do well in tests, but not so well when the attack patterns shift. This phenomenon is called concept drift, and it can be used to conclude that cybersecurity models are not static and are rather required to be monitored and updated regularly (Almarhabi & Elemery, 2025).

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A large number of cybersecurity studies show high model accuracy. But accuracy is not sufficient to determine usefulness of a system. Cybersecurity datasets can have an imbalance, with many more normal events than attacks or vice versa. A model can thus be both very accurate and not identify rare but serious threats. F1-Score, False Positive Rate and False Negative Rate should also be considered. False positives can be time consuming for analysts and false negatives can be dangerous as they allow serious attacks to go undetected. Therefore, a helpful cybersecurity risk model should take into account the expense as well as severity of each kind of mistake, not just the general accuracy of the system (Mohale & Obagbuwa, 2025; Prasad et al., 2026).

A big worry is the opacity of the complicated machine learning models. Although deep-learning and ensemble models can make good predictions, they might not be understood by the user in how they make a decision. This is what is known as the black-box problem. In the realm of cybersecurity, an unforeseen prediction might be hard to believe. Security analysts must be able to understand the rationale behind a system identifying an event as a threat. They also might have to justify their actions to managers, auditors or regulators. XAI can enhance the explainability of IDS systems by providing insights into the impact of various features on a prediction. The most common methods include SHAP, LIME, decision trees, and rule-based methods. But current research has issues with slow explanation processes, little real-time testing, limited scalability, and no standard ways of evaluating (Mohale & Obagbuwa, 2025).

SHAP and LIME are two popular explanation techniques. SHAP indicates the contribution of each feature towards the prediction. It can be used to explain a single result, or to give an overall picture of the behavior of the model. LIME normally explains one prediction by creating a simpler model around that case. Both approaches can be useful to analysts to understand the rationale behind the risk score. But, they have different functions. SHAP can sometimes be more consistent, but may take longer to process. While LIME might be more efficient in explaining locally, its results can vary depending on the samples and settings. Both techniques can be used to obtain global and local views of a prediction, but they can also add to the complexity of a system (Mohale & Obagbuwa, 2025; Prasad et al., 2026).

Distributed denial-of-service attacks have been detected using a convolutional network with SHAP and LIME. The model was able to obtain 94% accuracy, 93% precision, 95% recall and 94% F1-score. To identify the main features that influenced the model as a whole, SHAP was used and to explain individual predictions, LIME was used. Together, this was a strength as it gave both global and local explanations. But the model was primarily based on one attack type and on NSL-KDD dataset. This data set is commonly used, but is not a complete representation of the current network conditions. Additional testing should therefore be performed with up-to-date datasets and various types of cyberattacks (Salloum & Norozpour, 2025).

Machine learning and explainability have also been integrated in Smart Sustainable Cybersecurity model. The model was highly accurate in detecting threats and was able to explain individual classifications with LIME. Results indicated that predictions could be made that were interpretable without significantly degrading model performance. But, it was tested in a laboratory setting. The results do not prove that it would be as effective in a real network. Real systems feature encrypted traffic, partial labelling, evolving user behavior and new attack techniques. These conditions can lead to a decrease in the accuracy and explanation quality of the model (Almheiri et al., 2025).

The concept of explainability has also been used for malicious URL detection. Random Forest, Support Vector Machine, XGBoost, multilayer perceptron and deep-learning models were compared. SHAP and LIME explanations revealed the impact of URL length, domain structure, query content and character patterns on the predictions. A key discovery was that a more complicated deep-learning model did not necessarily outperform simpler models. It calls into question the notion that more complex models are necessarily more effective. When a simpler model can achieve the same or better performance, has less computational requirements and is easier to explain, then it may be more appropriate (Nair, 2024).

There is more sound technical work in the area of zero-day malware detection. One model fused the cross-attention features with SHAP, LIME, and Integrated Gradients. It achieved accuracy of over 99% on two malware datasets and demonstrated good results on unseen malware families. Feature importance was compared from multiple perspectives due to the use of multiple explanation methods. This is an asset as only one explanation approach may be sufficient to validate a finding. But results that are very high should be taken with

a pinch of salt. The behavior of malware can evolve rapidly and attackers can alter the files to fool both the prediction model and the explanation method. Real world and adversarial testing is thus required (Aljarrah et al., 2026).

Cybersecurity awareness studies also confirm the need for clear and actionable information. Research with University students reveals that knowledge is not always followed by safe actions. Where students have demonstrated vulnerabilities include browser security, phishing, social networking, and general cybersecurity practices (Alharbi & Tassaddiq, 2021). Lack of knowledge has also been reported on phishing, social engineering and secure Internet connections. There was a positive trend among students who had been educated on cybersecurity overall. These results indicate that a warning about a risky activity is not enough. They also must be given an easy-to-understand explanation of the threat and what to do about it (Erendor & Yildirim, 2022). Among the university students, it was found that the factors affecting cybersecurity awareness are password security, browser security, and social-media activity (Alqahtani, 2022). There is also a relationship between cybersecurity knowledge and password security and awareness, and users' attitudes may impact on the possibility of their knowledge leading to safer behaviors (Ahamed et al., 2024). Based on these findings, it is recommended that the security recommendations should be in line with the weakness identified by the system. If the risk is due to a weak password, for instance, then the recommendation should differ from if the risk is due to unsafe browsing or suspicious network activity. The general advice may not capture the real motivator for the predicted risk.

Security training research also indicates that a single suggestion might not be appropriate for all users. The results also revealed that there is a gap in cybersecurity awareness between faculty and staff, indicating that user roles can affect their security needs (Hobbs, 2023). Raising awareness can also not be enough to change behavior. One time security awareness activities may not be as effective as security championships, regular communication, feedback and role-related messages (Alshaikh & Adamson, 2021). The use of continuous cybersecurity-awareness frameworks has also been proposed for higher education. These are frameworks which acknowledge that students, academics, administrators, managers and technical staff are all exposed to various threats and need different types of guidance (Armas & Taherdoost, 2025).

The reviewed studies indicate a definite advancement in quantitative risk calculation, machine-learning prediction and explainability. These areas are however often

investigated separately. Quantitative risk models calculate risk and choose controls, however they might not provide explanations in the form of machine learning. Machine-learning models predict attacks, but they tend to end with a class, a probability or a risk score. Explain predictions, but many systems are not able to translate explanations into clear security actions. Awareness studies tell users what to do, but it is not usually on the basis of a particular machine-learning prediction.

This split poses a significant research gap. A good cyber security framework shouldn't just declare that an event is high risk. It should detail the primary reasons for the prediction and suggest actions consistent with the reasons. Several of the existing studies offer helpful components of this process, but few offer risk prediction, explainability, and risk-aware recommendations in a single framework. To fill this gap, the proposed CyberRiskXAI framework seeks to bridge the divide between machine-learning risk prediction and explanation, along with concrete recommendations. Its effectiveness can be evaluated based on prediction precision, false positive and false negative rates, explanation quality, processing time, relevance of recommendations and usefulness to cybersecurity professionals.

2.3 Problem Analysis

It was found from the literature reviewed that there are still some key gaps in existing cybersecurity risk assessment approaches. The traditional strategies rely on the expert judgement and the risk categories as low, medium and high. While easy to understand, these methods may not be as receptive to rapidly evolving cyber threats. Their findings are also dependent on the quality of the information on threats, vulnerabilities and controls in place. If the information is only partially available or if it is only estimated, the final risk score may not reflect the true situation of the organization (Sahinoglu, 2024).

Machine learning can help in this regard by processing vast quantities of cybersecurity data and recognizing patterns that may otherwise go unnoticed. Yet, numerous machine-learning research projects are limited to just identifying attacks and not assessing their broader impact. A model can determine if an event is malicious, but it may not be able to determine the severity of the event. This involves which assets are impacted or what damage can be done. The same attack can result in varying degrees of risk. The extent of the difference is dependant on the value of the system, the nature of the data and the

existing controls. Consequently, an attack label or probability score alone may not be sufficient for sound risk-based decision making (Abbas et al., 2025; Almarhabi & Elemery, 2025).

Another big challenge is data quality. Several cyber security models are trained and tested with benchmark or synthetic datasets. Such datasets are suitable for research, but do not adequately reflect real network environments. Real data can include unknown attacks, changing behavior, new devices and encrypted traffic. Hence, a model with a high test accuracy might not behave the same way when deployed. Cyber security data is also generally skewed, since there are many more normal events than attacks. In such situations, a model can show high accuracy, but still miss out on some rare and serious threats. Precision, recall, false-positive rates and false-negative rates should therefore be taken into account in addition to accuracy (Mohale & Obagbuwa, 2025; Prasad et al., 2026).

Another issue is that many sophisticated machine learning models are "black boxes" in that they are difficult to understand. Deep-learning and ensemble models can yield great results, but can be hard to explain. This lack of transparency can lower the level of trust, as the analysts want to know why an event was rated as "dangerous." They might also have to explain why they blocked access, isolated a device, or investigated a user. If there is no explanation, it is not easy to verify if the model has been using some relevant security features, or if it has been using irrelevant data patterns (Mohale & Obagbuwa, 2025). Explainable AI techniques like SHAP and LIME can be of assistance in demonstrating which features impacted a prediction. These methods, however, also have their drawbacks. SHAP can be time-consuming, particularly when dealing with large datasets and complex models. While LIME can explain individual predictions, the results can vary with the settings. Explanations might also be overly technical or hard to follow for security professionals. A feature-importance chart does not necessarily tell you what is going on, why it is important, or what to do next. Thus, an explanation should be precise, permanent, comprehensible and useful in real security work (Ashfaq & Chowdhury, 2023; Eriksson & Grov, 2022).

There is also a disconnect between prediction, explanation and action in the literature. Many systems can detect a threat and tell you what is the key component to the result. But they tend to just end their sentences without giving an adequate answer. General advice

like changing passwords or using multi-factor authentication may not be appropriate to the specific source of risk. For instance, if the risk is associated with an unusual login pattern, it should be handled differently than if it's due to outdated software or suspicious network traffic. Recommendations should thus be related to the level of risk predicted, and the factors that underlie this risk (Gamboa Gamarra, 2026; Rahman et al., 2025).

These weaknesses can have serious consequences. False positives: More work for the analyst and can create alert fatigue. False negatives can lead to the failure to detect harmful attacks. Confusing explanations can diminish trust in automated systems, and vague recommendations can result in sub-optimal or unnecessary security measures. In general, the literature indicates that the study of risk prediction, model explanation, and security recommendations tends to be done separately. This generates an obvious need for research linking these areas. Hence, the proposed CyberRiskXAI framework is valid as the cybersecurity professionals not only require the accurate prediction of cybersecurity risks but also an explanation of the result and guidance that corresponds to the risk identified.

2.4 Summary

This chapter summarizes the state-of-the-art work on Cybersecurity Risk Assessment, Risk Prediction using Machine Learning, Explainable Artificial Intelligence and Risk-aware Recommendation. As per the literature, machine learning can help in faster and more accurate cyber threat detection and risk prediction. However, many models rely on benchmark datasets and may not work as well in real world settings. The review also identified that complex models are often used as black boxes and their predictions are not easy to understand and trust. Some techniques like SHAP and LIME can add transparency, but their explanations are slow, unstable, and can be too technical for practical implementation (Mohale & Obagbuwa, 2025; Prasad et al., 2026). A second important result is that most studies only address prediction and/or explanation and/or recommendation, but not all three. This means that security professionals can get a risk score for no apparent reason and have no way to react. This gap is a justification for the proposed CyberRiskXAI framework. The research is focused on bridging the gap between cybersecurity risk prediction and lucid explanation and actionable recommendations aligned with the identified risk. The methodology adopted for the design, development and evaluation of the proposed framework is presented in the next chapter.

Chapter 3

Proposed Model

3.1 Introduction

In this chapter, the proposed model, "CyberRiskXAI: An Explainable Machine Learning Framework for Cybersecurity Risk Prediction and Risk-Aware Recommendations" is introduced. The model centers on cybersecurity threats stemming from users' awareness, behavior and everyday usage of digital systems. These risks are significant for people and universities, private companies and government institutions and large enterprises. When many users are working on connected computers and shared digital services in an organization, there is the potential for one unsafe action to put data, systems, and other users at risk. In this chapter, the key architecture, function and scope of the proposed CyberRiskXAI framework are introduced.

3.2 Feasibility Analysis

The feasibility analysis will check the viability of the proposed CyberRiskXAI framework for implementation in the real world. It takes into account, technical, operational, economic, schedule and enterprise feasibility.

Technical Feasibility:

Technically feasible due to the use of Python, Google Colab, machine learning libraries and Streamlit. These tools are broadly accessible and appropriate for model development and Web deployment. System can be accessed by users with a computer, Internet connection, and web browser.

Operational Feasibility:

Technical and non-technical users will find the system easy to use. People can fill in a questionnaire and get a risk assessment, explanation and recommendation. Organizations can also see the overall risk condition of their employees, students or users.

Economic Feasibility:

The framework is financially viable because most of the tools used are freely available and open source. The basic system can be built and implemented in a cost-effective manner. Secure hosting, databases and access control are additional costs that might be needed in

large organizations.

Schedule Feasibility:

The framework can be developed during the research period because it is divided into sequential stages. This includes data preparation, model development, integration of Explainable AI, website development, testing, and documentation.

In general, the proposed framework would be viable as it leverages existing technology, would be relatively inexpensive, and would be able to accommodate both personal and organizational cybersecurity risk assessment.

3.3 Requirement Analysis

The requirements of the CyberRiskXAI framework encompass the fundamental processing, user interaction, data management, and output capabilities. The system should be able to receive a cybersecurity survey, clean and encrypt the data, assign a safety score and risk score, and categorize users into Low Risk, Medium Risk, or High Risk. It should deliver the trained machine learning model to new responses, explain the primary drivers behind each prediction using Explainable Artificial Intelligence and be able to suggest risk-aware recommendations based on poor cybersecurity practices. Individual users should be able to complete the survey, submit their answers, look at their risk score, view the reasoning behind the prediction, and be given personal recommendations. Organizational users should be capable of uploading data for several users, view the overall risk situation, identify common weaknesses and apply the results to awareness and training programs. The system should also validate data entered, accommodate missing and incorrect data, record data in a structured format, safeguard personal and organizational information, and provide for future changes to the model, survey questions and recommendations. The output should contain the predicted risk level, safety and risk scores, explanation of important factors, suitable recommendations and enterprise level summary of percentage of low, medium and high-risk users.

3.4 Research Methodology

The CyberRiskXAI framework is interdisciplinary, merging cyber security, machine learning, Explainable Artificial Intelligence, and software engineering. The research starts

by conducting a structured survey. The survey gathers data concerning users' cybersecurity awareness and their daily cyber activities. The questions ask about password re-use, two factor authentication, phishing knowledge, unknown links, sharing of passwords, public wi-fi, antivirus software, privacy, secure sites, password management, and software installation. Shared systems, employee awareness, access control, software updates, incident reporting, and organizational data protection are also taken into account in the framework for enterprise-level assessment. These are the answers to evaluating cybersecurity risk in persons and groups of persons. Once data has been collected, the data set is cleaned and prepared for analysis. Extra columns, duplicate data, blank data, spelling issues, and answer formats are evaluated. The responses to the text based survey are next converted to numbers. More values are awarded for safe cybersecurity practices with less values for unsafe practices. Each respondent receives a Cyber Safety Score, and a Cyber Risk Score, based on these values. Safety Score categorises the users in 3 groups: Low Risk, Medium Risk and High Risk. Low Risk users typically adhere to safe practices when using the Internet. High Risk users are more likely to engage in unsafe behaviour. Medium Risk Users: This user understands the risk of cybersecurity, but does not always practice safe habits. Individual results are aggregated for enterprise level analysis. This enables an organization to see the proportion of users per risk group and what the typical cybersecurity vulnerabilities are. Training and testing data sets are generated by stratified splitting of the prepared data set. This approach maintains the percentage of all three risk categories the same for each set. The models are trained on the training set and tested on the testing set to evaluate the models on unseen data. As it is a structured dataset and derived from surveys, supervised machine learning is more appropriate than deep learning. Deep learning was not chosen because it typically demands a larger data set, more computing power, and is often not easy to explain. In this research, four supervised machine learning models are tested namely Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression, and Random Forest. Balanced Decision Tree is incorporated as it can deal with class imbalance and generate comprehensible decision rules. Comparable probability model Gaussian Naive Bayes is employed. Logistic Regression is appropriate for classification and provides transparent relations between input features and risk classes. Random Forest is an ensemble of multiple decision trees and can be used to overcome overfitting. The same data set is used for training and testing of all models. They are evaluated based on accuracy, precision, recall, F1-score, classification reports and confusion matrices. Accuracy indicates the overall prediction accuracy. Precision is the

accuracy with which the class labels are predicted. Recall is indicative of the proportion of actual users in each risk group correctly identified by the model. The F1-score is a combination of precision and recall. The confusion matrix illustrates correct and incorrect predictions for each class. The model with the best performance, class level results and interpretability is selected. Explainable AI is then incorporated, which makes the predictions understandable. SHAP provides an overall importance of the Cyber Security features while LIME provides an explanation on the prediction made by each individual feature. The identified weak behaviours are then leveraged to create appropriate cybersecurity recommendations.

Lastly, the chosen model, XAI techniques, and recommendation system are implemented in a Streamlit web app. The platform supports receiving surveyed answers, predicting the level of risk, explaining the results, offering recommendations, and enterprise-level summary.

3.5 System Design and Architecture

CyberRiskXAI system is designed following the modular structure, consisting of two main parts – model development environment and web-based application environment. The machine learning model is created, trained, tested and saved on Google Colab. New surveys are evaluated with the saved model and the cybersecurity risk results are displayed on the Streamlit website. The separation of these two parts simplifies the development, testing, maintenance and updating process.

Model Development Architecture

Developing models in Google Colab. It has the parts where the dataset is prepared, trained, tested, evaluated and model are saved. The information is first uploaded to the notebook. Preprocessing: Cleaning Data, encoding survey responses to numerical values, creating the required features, and creating the desired Risk Labels. Processed data is then passed along to the model training part. Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression and Random Forest classifiers are trained using the same prepped data set. The models are compared using standard classification measures, as part of the evaluation component. Comparing the results, the most appropriate model is selected and saved as trained model file. The order of the features, the value mapping as well as the preprocessing structure are also preserved, to allow the website to process new responses in the same manner.

The flow of the development of the model is:

Survey Dataset $\Rightarrow$ Data Preparation $\Rightarrow$ Feature Encoding $\Rightarrow$ Risk Labelling $\Rightarrow$ Model Training $\Rightarrow$ Model Testing $\Rightarrow$ Model Selection $\Rightarrow$ Saved Model

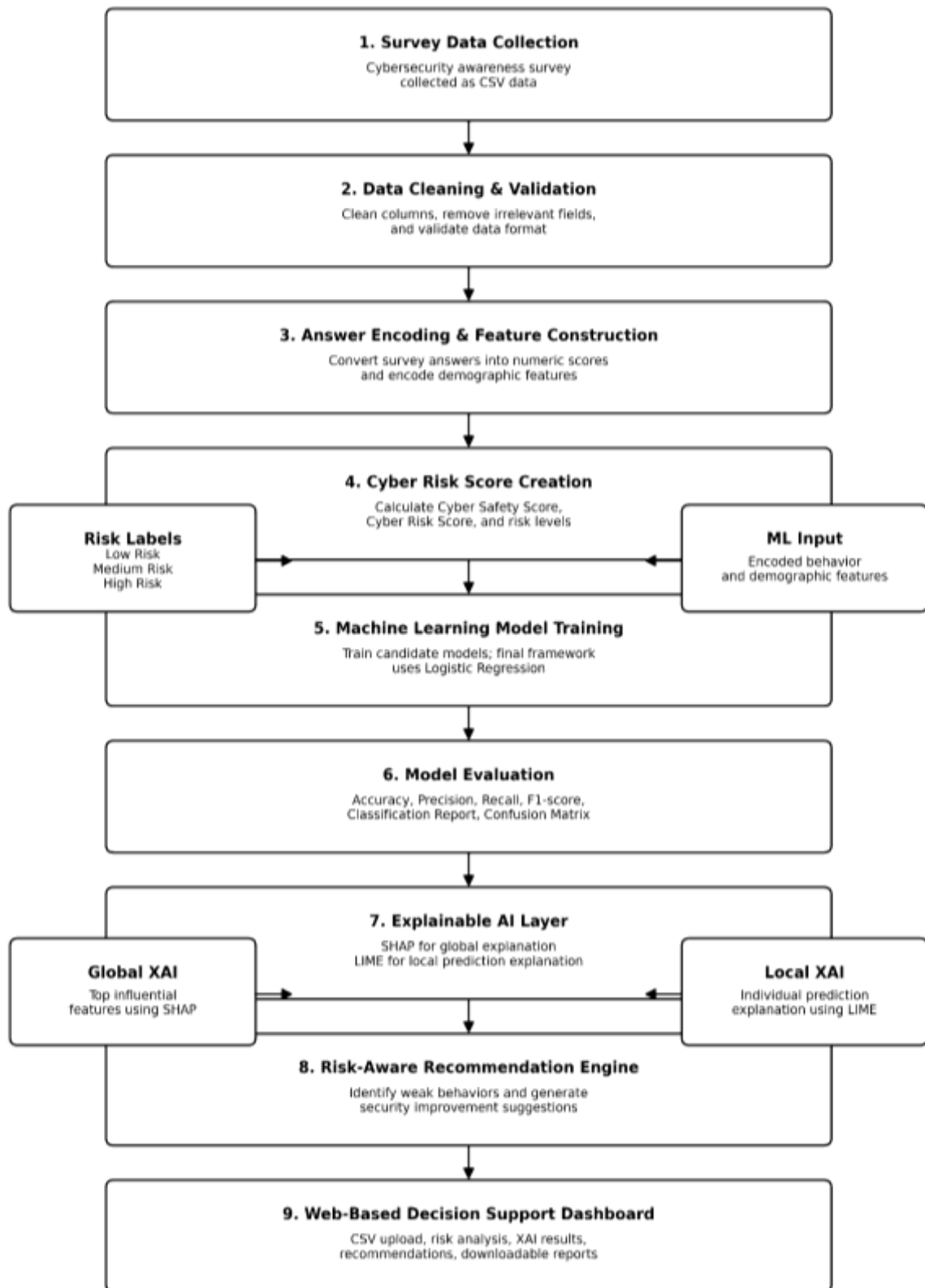


Figure 2: CyberRiskXAI Architecture

Web Application Architecture

The web app is created using streamlit. The running part of the CyberRiskXAI framework. The website is not retrained with the machine learning models. Rather it loads the model that has already been trained and saved from the Google Colab development process.

The presentation layer is the layer where users and the organization are interacting with the system. A user or an organization can complete the cybersecurity survey, which allows for multiple users, students, or system users to complete it. The responses can be grouped by enterprise or department for organizational analysis.

The main operations of the system are controlled by the application layer. It takes the data submitted and verifies if the necessary fields are filled and returns the responses to the data transformation component. This component consists of the same numbers and features as used in model training. It is used to ensure that the site that is entered will be in the correct format expected by the saved model. The transformed data is then fed into the prediction component where a risk category is determined. The records are classified as Low Risk, Medium Risk and High Risk. The system combines the predictions of several users when used in an enterprise. It then determines the number of users in each risk group and gives percentages. XAI will then provide an explanation of the prediction result. SHAP can visualize which are the most important risk factors for the overall dataset and LIME can explain a specific prediction. The recommendation portion of the report identifies the weak behaviors and the identified risk and offers the relevant cybersecurity recommendations. The dashboard is used to present the results to the output layer. It may contain anticipated risk category, vital contributing factors, risk-aware recommendations, and risk distribution charts and enterprise level summaries. This makes it possible for technical and non-technical users to comprehend the cybersecurity state of an individual or organization.

The flow of the web application:

Input: Survey/Enterprise Input → Input Validation → Data Transformation → Saved Model → Risk Prediction → XAI Explanation → Recommendation Generation → Dashboard Output

The features of integration and security design are included. The elements of integration and security design are covered. The model is saved and the preprocessing structure is created to connect Google Colab with the Streamlit website. To update the model, a new trained file was created in Google Colab, which could be used in place of the existing trained file in the website. This way the operation of the system and the development of the model can be decoupled. Data security and privacy are also taken into account in the

system design. It is important to collect only survey information that is necessary. Input should be checked before processing and reports of the organization should be protected from unauthorized use. The system may feature user authentication, role-based access management, encrypted communications, secure data storage, and restricted access to enterprise outcomes to support enterprise deployment.

The overall architecture is the combination of the model created in Google Colab and the Streamlit web platform. Model development is carried out in Google Colab, and the website handles user interaction, prediction, explanation, and recommendations and the presentation of the results.

3.6 Implementation and Simulation

The implementation of CyberRiskXAI was completed in two connected stages. The first stage focused on data preparation, model training, evaluation, and model selection in Google Colab. The second stage focused on connecting the selected model with Explainable Artificial Intelligence methods, a recommendation module, and a Streamlit web application. The complete system was then tested through individual and enterprise-level simulations.

Implementation

The implementation began by importing the cybersecurity survey dataset into Google Colab. The dataset contained demographic information and answers related to users' cybersecurity awareness and behaviour. The questions covered password reuse, two-factor authentication, phishing awareness, password sharing, public Wi-Fi use, antivirus software, privacy practices, secure websites, password management, software authenticity, and other online security habits.

The dataset was checked before model development. Unnecessary columns were removed, and duplicate, incomplete, and inconsistent records were reviewed. Column names were shortened, spelling errors were corrected, and text formats were made consistent. This ensured that similar answers were treated in the same way during numerical conversion. The survey answers were then encoded using a three-level scoring system. Unsafe behaviour received a score of 1, partly safe or inconsistent behaviour received a score of 2, and safe behaviour received a score of 3. For example, password reuse received a lower

score, while using unique passwords received a higher score. Demographic features such as gender, age, department, and year of study were converted through one-hot encoding. The scores from 15 cybersecurity questions were added to create a Cyber Safety Score. The possible score ranged from 15 to 45. Higher scores represented safer cybersecurity behaviour, while lower scores represented weaker behaviour. The respondents were classified into three groups: High Risk for scores from 15 to 25, Medium Risk for scores from 26 to 35, and Low Risk for scores from 36 to 45. These risk labels were also converted into numerical classes for machine learning. Cyber Safety Score and Cyber Risk Score were removed from the input features because they were used to create the target labels. This helped prevent data leakage. The dataset was divided into 80% training data and 20% testing data using stratified sampling. This maintained the proportion of Low Risk, Medium Risk, and High Risk records in both sets. Four machine learning models were trained using the same data: Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression, and Random Forest. The models were evaluated using accuracy, precision, recall, F1-score, classification reports, and confusion matrices. Logistic Regression was selected as the final model based on the comparative results. The selected model was saved with its feature order, numerical mappings, demographic encoding structure, and class labels. These saved components ensured that new survey responses were processed in the same way as the training data. The Streamlit application was then developed as the user-facing part of the system. The application loads the trained model without retraining it. Individual users can complete the survey and receive a predicted risk category. Organizations can upload multiple survey records and view the number and percentage of users in each risk group. SHAP and LIME were connected to the prediction process. SHAP was used to show the overall influence of cybersecurity features, while LIME was used to explain individual predictions. The identified weak behaviours were passed to the recommendation module, which provided suitable cybersecurity advice. The final dashboard displayed the predicted risk level, main contributing factors, recommendations, charts, and enterprise-level summaries.

Simulation

The completed system was tested using the reserved testing dataset and several cybersecurity behaviour scenarios. Safe scenarios included unique passwords, regular use of two-factor authentication, safe browsing, and trusted software. Moderate scenarios included both safe and unsafe behaviours. Unsafe scenarios included password reuse,

suspicious links, public Wi-Fi for sensitive activities, weak account protection, and untrusted software. These scenarios were submitted through the Streamlit application. The predicted risk levels were checked to confirm that the system produced reasonable Low Risk, Medium Risk, and High Risk results. The XAI explanations were reviewed to ensure that the displayed factors matched the submitted answers. The recommendations were also checked to confirm that they addressed the detected weaknesses. The enterprise function was tested by processing multiple survey records together. The system produced individual predictions, an overall risk summary, risk percentages, and common cybersecurity weaknesses. Finally, the full workflow was tested from survey submission to dashboard output. The simulation showed that the model trained in Google Colab worked correctly within the deployed Streamlit application.

3.7 Tools and Technologies Used

The CyberRiskXAI framework was developed using Python, Google Colab, Pandas, NumPy, Scikit-learn, SHAP, LIME, Streamlit, CSV, and Streamlit Cloud.

Python was used as the main programming language because it supports data processing, machine learning, Explainable AI, and web development. Google Colab was used to upload, clean, encode, train, and test the survey dataset without requiring a high-performance local computer. Pandas was used for data cleaning, missing-value handling, numerical mapping, score creation, and CSV processing. NumPy supported mathematical and array-based operations. Scikit-learn was used to build and evaluate Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression, and Random Forest models. It was also used for train-test splitting, encoding, prediction, accuracy, precision, recall, F1-score, classification reports, and confusion matrices. SHAP was used to show the overall influence of cybersecurity behaviours on risk predictions. LIME was used to explain individual predictions and identify the survey answers that increased or reduced a user's risk. Streamlit was used to create the CyberRiskXAI web application. The application accepts survey responses, processes the data, loads the trained model, predicts the risk level, displays explanations, and provides recommendations. CSV files were used to store and upload survey records because they are simple and widely supported.

Finally, the application was deployed through Streamlit Cloud and accessed using a web browser. These tools supported the complete workflow, from data preparation and model

development to Explainable AI, web deployment, and enterprise-level cybersecurity risk assessment.

3.8 Summary

This chapter introduced the proposed CyberRiskXAI framework to predict cybersecurity risks and generate cybersecurity recommendation based on the predicted risks. It explained the feasibility, requirements of the system, methodology of the research and architecture of the overall system. The chapter also described the processes for cleaning, normalizing the data from the surveys and mapping them with numerical values and categorizing the users as Low Risk, Medium Risk and High Risk. The machine learning models were trained and compared in Google Colab and the results were explained using SHAP and LIME. The model selected was then used in a Streamlit web app for individual and enterprise assessment. The framework also gives recommendations on the basis of cyber security weaknesses noted in the responses to the surveys. The chapter provided a general overview of CyberRiskXAI, a combination of machine learning, explainable AI and web-based implementation to solve the problem of human-centric cybersecurity risk. The implementation process, testing procedures and evaluation results of the system's performance are presented in the next chapter, followed by a discussion of the system's performance.

Chapter 4

Implementation and Testing

4.1 Introduction

The proposed CyberRiskXAI framework was introduced in the previous chapter. It described research methodology, system design, data processing procedure, machine learning models, Explainable AI methods, and web-based system design. It also outlined the methodology to classify the survey responses into three classes (Low Risk, Medium Risk, and High Risk) and offered appropriate cybersecurity recommendations. The implementation and testing of the CyberRiskXAI system is presented in this chapter. It

provides an explanation of the data in the survey and the training and testing of the machine learning models. It also details the integration of the chosen model with SHAP, LIME, the recommendation module, and the web app. The system setup, software tools, testing process, and performance measures are discussed in this chapter. The models' results are then presented and analyzed. Accuracy, precision, recall, F1 score, classification report and confusion matrix are discussed. The chapter also explores the validity of the final system for individual and enterprise cybersecurity risk assessment. This chapter provides a general overview of the practical development of the proposed framework. It also assesses the feasibility of achieving the research goals and delivering clear, reliable and useful cybersecurity risk results from the system.

4.2 System Setup and Environment

The CyberRiskXAI system was implemented and tested using a cloud-based development environment. Data preparation, model training, testing and evaluation was done using Google Colab. This obviated the necessity of a high-speed local computer. The notebook was accessed and experiments conducted on a standard computer with internet access and a web browser. Google Colab computing resources were appropriate for the size and composition of the survey data.

Python was used as the main programming language. Survey data was loaded, cleaned, transformed and managed using pandas. NumPy provided numerical operations and data processing. The data was split using scikit-learn, one-hot encoded, trained, predicted, and evaluated using scikit-learn. The machine learning models used were Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression and Random Forest. All models were trained and tested on the same data and on the same test set, to ensure a fair comparison. The data was saved and analyzed in the CSV file format. The split was 80/20 for training and testing, respectively. Both sets of records had the same proportions of Low Risk, Medium Risk and High Risk records as stratified sampling. The accuracy, precision, recall, F1-score, confusion matrix and classification reports were used to evaluate the models. Explainable Artificial Intelligence was done using the tools: SHAP and LIME. The features with the greatest overall impact on the predictions were identified using SHAP. Individual risk predictions were explained using LIME. Streamlit was used to develop the web-based application. The trained model was stored and loaded from Google Colab to the Streamlit

application. The application received the answers to the survey, calculated the values, gave risk assessments, created explanations, and gave risk-aware recommendations.

The system was deployed via Streamlit Cloud and accessed via a standard web browser. This configuration enabled individual and enterprise level testing. It was also easy to reproduce as the main tools and libraries used were free, widely available and compatible with Python.

4.3 Implementation of the System

The CyberRiskXAI system was deployed in two consecutive phases. The first step was to prepare the data and develop the machine learning models in Google Colab. The second stage involved the integration of the chosen model with Explainable AI, recommendations, and a Streamlit web application. This architecture separated model development from the running web system.

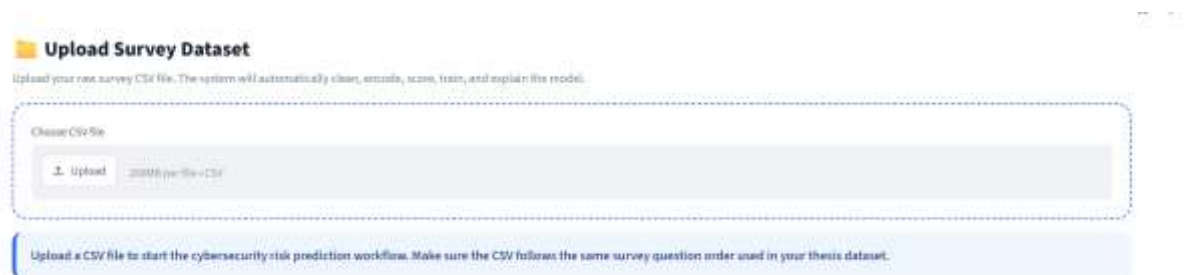


Figure 3: CyberRiskXAI User Dashboard page

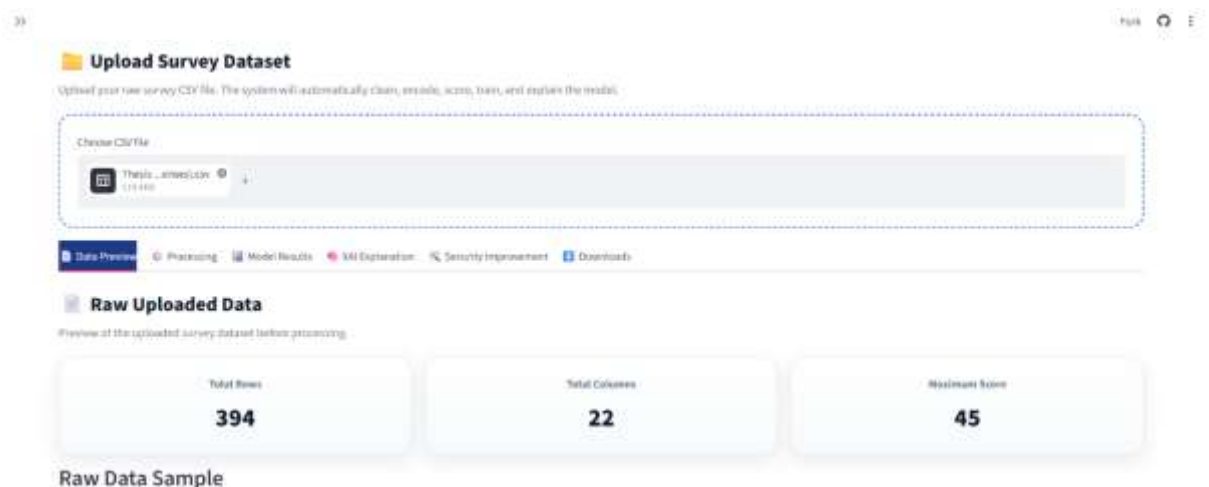


Figure 4: Data Information page

Upload Survey Dataset

Upload your raw survey CSV file. The system will automatically clean, encode, score, train, and explain the model.

Choose CSV file

Thesis...csv 118 KB

Data Preview Processing Model Results XAI Explanation Security Improvement Downloads

Data Cleaning, Encoding, and Risk Scoring

The system converts text answers into numeric cybersecurity behavior scores.

1 = risky behavior, 2 = moderate behavior, 3 = safer behavior.

Cleaned Dataset

Figure 5: Data Cleaning, Encoding, risk scoring page

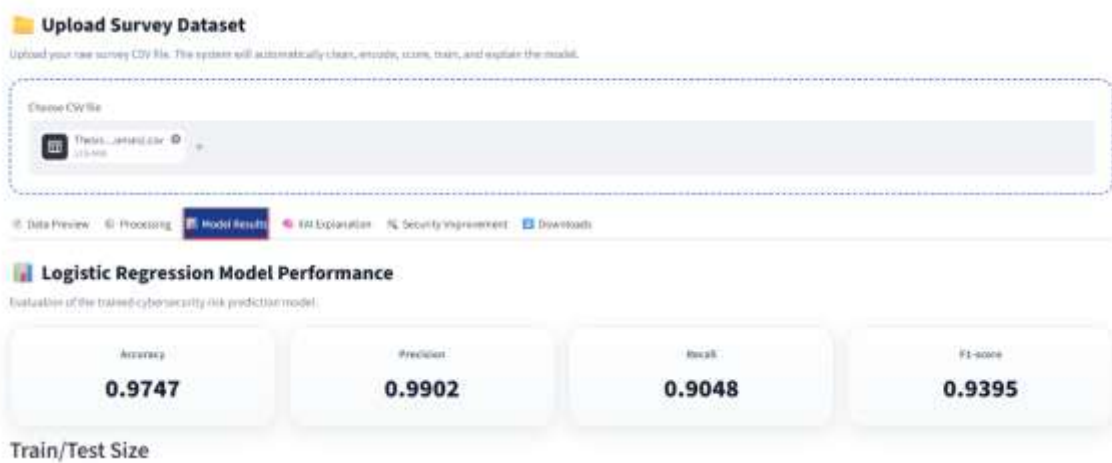


Figure 6: Model Performance page

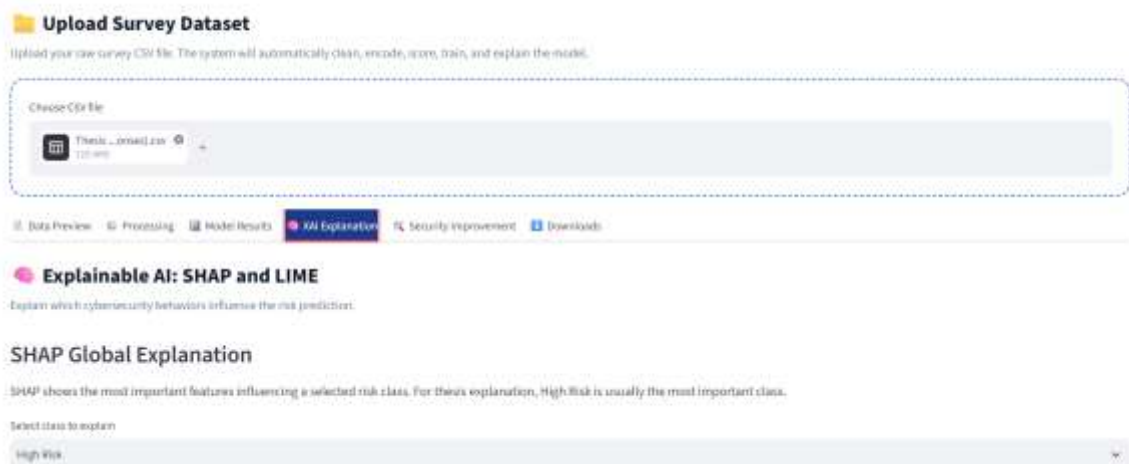


Figure 7: Explainable AI page

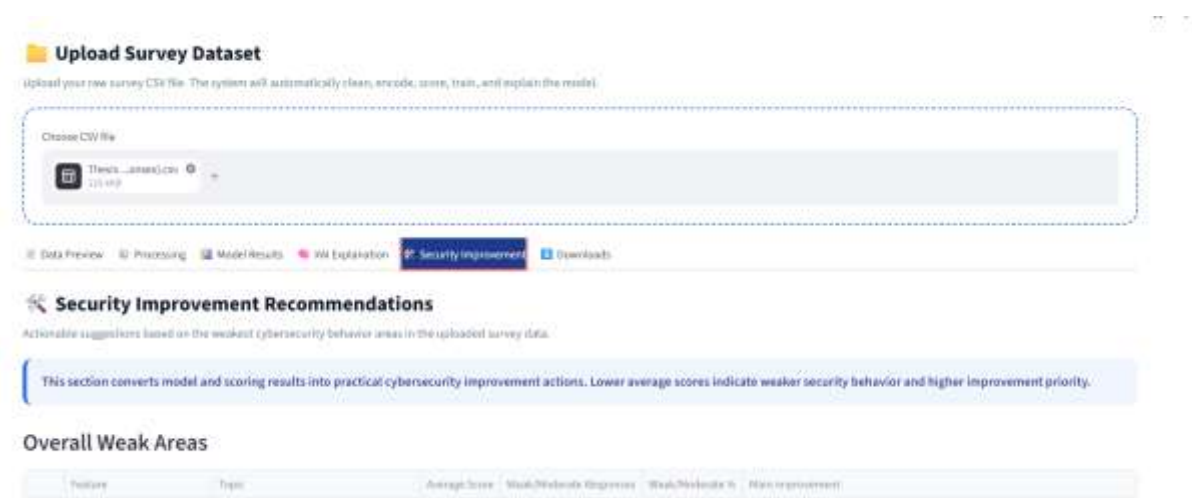


Figure 8: Security Improvement and Recommendation

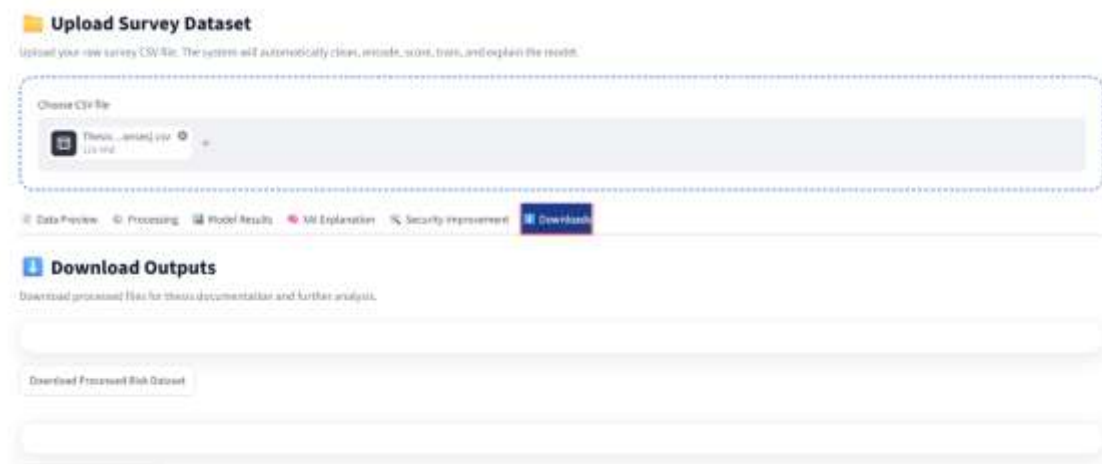


Figure 9: Download reports page

The first step of the implementation was to load the cybersecurity survey data into Google Colab. Unnecessary columns were eliminated. Duplicate, incomplete and inconsistent records were rectified. The names of columns and text formatting were also fixed. The answers to the survey were then coded into numbers. If a child was behaving in an unsafe way, it was given a score of 1. Partly safe or inconsistent behaviour was scored as 2. Safe behaviour was given a score of 3. One-hot encoding was performed on the demographic data, including gender, age, department, and year of study. Security responses encoded were used to compute a Cyber Safety Score. The range of scores was from 15 to 45. Higher scores indicated safer behaviour. Weaker cybersecurity behaviors were indicated by a lower score. A Cyber Risk Score was also created from the safety score. The respondents were then classified as High Risk, Medium Risk, or Low Risk. These were not considered

as the features of the model, Cyber Safety Score and Cyber Risk Score. This avoided data leakage since these scores were already used to generate the target labels. The prepared data set was split into 80% training data and 20% test data. The proportion of Low Risk, Medium Risk and High-Risk records has been maintained as similar in both sets by stratified splitting. Google Colab was used to implement four supervised machine learning models. They were Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression, and Random Forest. The same input features, training set and testing set were used for each model. This provided a fair and consistent comparison. The models produced predictions for the unseen testing data. The most appropriate model was then saved for use in the CyberRiskXAI web application after the evaluation process. The order of features, the mapping of answers to features, the structure of the demographics, and the labels of the classes were also saved. This guaranteed that new survey responses were read in the same format as the training data. The user-facing part of the system, Streamlit application, was created. The trained model is loaded into the application and is not retrained in normal use. The application allows for individual and enterprise assessment. The survey can be completed by an individual user, and the risk level predicted. An organization can upload several survey records and get a summary of risk. Upon submission of data, the system verifies if the necessary fields are filled. It then remaps the answers with the same mappings and feature structure as used in model training. The modified data are sent to the saved model. The model predicts Low Risk, Medium Risk or High Risk. The system processes each record separately in enterprise assessment and calculates the number and percentage of users in each risk group.

The Explainable AI module was hooked into the prediction process. The overall impact of cybersecurity behaviours was demonstrated using SHAP. Individual predictions were explained using LIME. The weak behaviours identified were then transmitted to the recommendation module. This module generated suitable cybersecurity advice. The final dashboard included the level of risk, key factors, recommendations, charts, and enterprise-level summaries. At the research prototype level, security was limited to gathering only the necessary survey data and ensuring that the data was accurate before it was used. The protection of organizational reports was also considered in the system design. If it's to be deployed in the enterprise, more controls would be needed, including user authentication, role-based access, encryption, and secure data storage.

4.4 Evaluation and Testing

The CyberRiskXAI system was tested at two levels. The first level was an evaluation of the machine learning models. The second level tested the full Web application. This was done to ensure the model gave accurate predictions and all the system modules worked correctly. The data set was split into training data of 80% and testing data of 20%. To ensure that the number of records in each set were approximately equal, stratified splitting was employed. Testing records were not utilized in the model training. Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression and Random Forest were trained and tested using the same sets of training and testing data. This allowed for a fair comparison. Cyber Safety Score and Cyber Risk Score were also excluded from the model inputs as they were previously used to develop the risk labels. This helped to prevent data leakage. Models were assessed by accuracy, macro precision, macro recall, macro F1 score, classification reports and confusion matrix. Accuracy: The percentage of all predictions that were correct. Precision is the percentage of times that a predicted risk class was accurate. Recall measures the accuracy with which the model was able to identify the actual users in each risk class. The F1-score is a balance between precision and recall. Macro averaging was employed as it provides equal weighting to Low Risk, Medium Risk and High Risk, regardless of the number of records in each class. Confusion matrix was employed to illustrate the correct and incorrect predictions for each class. The greater number of Medium Risk records in the dataset compared to Low Risk and High Risk records meant that accuracy was not used as the sole means of selecting the final model.

Model	Accuracy	Precision	Recall	F1-score
Balanced Decision Tree	89.87%	83.81%	68.04%	73.80%
Gaussian Naive Bayes	70.89%	61.85%	75.61%	60.51%
Logistic Regression	97.47%	99.02%	90.48%	93.95%
Random Forest	92.41%	97.22%	69.84%	78.55%

Table 1: Model Results

A Logistic Regression confusion matrix was also analyzed. There were 79 records in the testing set. 100% of the 6 actual Low Risk records were correctly predicted as Low Risk. 66 actual Medium Risk records were accurately forecasted as Medium Risk. Of the 7 actual High Risk records, 5 were accurately predicted as High Risk and 2 were predicted as Medium Risk. So, the model was correct in 77 predictions and wrong in 2 predictions. There was no record in the wrong class for either Low Risk or Medium Risk.

The Web application was validated with individual and enterprise level simulations. The scenarios of safe, moderate, and unsafe cybersecurity behavior were added to the Streamlit application. The levels of risk expected were checked to ensure that they matched the behaviour entered. The factors shown in the SHAP and LIME outputs were verified to ensure that they were related to the answers provided. The recommendations were also reviewed to make sure that they covered the identified weak behaviours.

Functional testing at the module level was conducted on input validation, answer encoding, feature arrangement, saved-model loading, risk prediction, XAI explanation, recommendation generation and dashboard display. Several survey records were used for the enterprise function test. The risk count, the percentage of each class, and the common cybersecurity risks were verified. The entire workflow was also tested from the survey submission to the final dashboard output.

Simple security measures on data validation and controlled data processing. Prediction was performed after the verification of missing or incorrect values. Only relevant information for the assessment was processed. It was not subjected to a formal penetration test as it was being developed as a research prototype. Before large scale enterprise deployment, a full security audit, increased authentication, encrypted storage, and role-based access would be required.

4.5 Results and Discussion

The results of the evaluation demonstrate significant differences between the four machine learning models. The best overall performance was obtained by Logistic Regression. It produced an accuracy of 97.47%, precision of 99.02%, recall of 90.48%, and an F1-score of 93.95%. The results indicate that the model predictions were very reliable and had a good precision and recall balance. It is also fair to compare the four models since the training and testing data was the same for all of them.

The confusion matrix provides a better insight into the Logistic Regression results. There

were 79 records in the test set. 100% of the 6 Low Risk records were accurately identified. All 66 Medium Risk records also were correctly classified. Of the 7 High Risk records, 5 were correctly classified and 2 were classified as Medium Risk. Hence, the model predicted correctly 77 times and incorrectly 2 times.

There was 100% recall among the Low Risk and Medium Risk classes. The recall rate for the High Risk class, however, was around 71.43% as it was able to correctly identify 5 of the 7 High Risk records. This is a significant restriction. The consequences of not identifying a High Risk user could be worse than incorrectly identifying a safer user as a High Risk user. The two errors were not extreme as the High Risk users were classified as Medium Risk instead of Low Risk. But the system needs to be further enhanced to be able to identify High Risk users more reliably.

The second highest accuracy was obtained by Random Forest with 92.41%. It also had a high precision of 97.22%. However, its recall was only 69.84%, and its F1-score was 78.55%. This indicates that it was sometimes right when a class was chosen, but that it could not pick out some of the true records from the various risk groups. It was not as good as Logistic Regression for the final system because of its lower recall.

The Balanced Decision Tree performed with an accuracy of 89.87%, precision of 83.81%, recall of 68.04% and an F1-score of 73.80%. It gave clear decision rules and was helpful in dealing with the unequal class distribution. However, it has a lower recall and F1-score, indicating that it failed to recall more records from the actual risk classes. It proved useful as a comparison model but was not chosen for deployment.

The Gaussian Naives Bayes had the poorest overall performance. Its accuracy was 70.89%, precision was 61.85%, recall was 75.61%, and F1-score was 60.51%. It had a higher recall than the Balanced Decision Tree and Random Forest. Its low precision and low F1-score indicates that many of its predictions were not reliable, however. A possible explanation is that Gaussian Naive Bayes assumes that the values of the inputs are normally distributed. Most of the data in the CyberRiskXAI dataset is survey category-encoded and binary demographic data. These values do not necessarily adhere to that assumption.

It is observed that Logistic Regression achieved better accuracy by 5.06 percentage points and F1 score by 15.40 percentage points compared to the Random Forest model. It also outperformed the Balanced Decision Tree and Gaussian Naive Bayes on four major evaluation metrics. The Logistic Regression model could separate the relationships

between the encoded cybersecurity behaviours and the risk categories effectively, suggesting that the model would be effective for this purpose.

The findings are consistent with the primary research goal of forecasting cybersecurity risk from survey responses due to human actions. The final model is able to classify the users with high overall performance into Low Risk, Medium Risk and High Risk groups. Logistic Regression is also appropriate for the proposed framework because it is easier to understand than many other, more complex, models. This helps to explain the use of SHAP and LIME to illustrate the impact of the various responses to the survey on the predicted risk. The model, XAI modules, recommendation process, and Streamlit interface were all created to be a single integrated system. In addition, individual survey responses were also able to be processed through the stages of data validation, transformation, prediction, explanation and recommendation in the system simulation. Enterprise-level summaries could also be processed using multiple records. The dashboard presented the risk distribution and common cybersecurity weaknesses. This means that the main functional objectives of the system were achieved at the research prototype level. The results should be interpreted with caution, however. The data set was fairly small and had a higher number of Medium Risk than Low Risk and High Risk records. Only 6 Low Risk and 7 High Risk records were used in the test set. Therefore, there could be significant variation in class-level performance if one or two predictions are incorrect. The large number of Medium Risk records also may contribute to the high accuracy. Stratified splitting minimized this problem, but not eliminated it.

The comparison that is currently being made is only for four models that were trained on the same data set. No experimental comparison was made with other models published in the literature. The results indicate that Logistic Regression is the best model in this research, but this does not imply that it will outperform all the available cybersecurity risk prediction techniques. More testing with larger sample size, more balanced risk classes, more external validation would provide more evidence.

Logistic Regression was chosen as the final model as it had the highest accuracy, precision, recall, and F1-score overall. It also enabled easy interpretation and integration with the CyberRiskXAI application. The results address the primary goals of the study, but additional data collection and enhanced High Risk detection is needed before the system can be deemed ready for actual use in a large enterprise.

4.6 Summary

In this chapter, the implementation, testing, and evaluation of the CyberRiskXAI system were introduced. All four machine learning models were trained with the same training data, and the same test data were used to compare all four models. Logistic Regression had the highest overall accuracy of 97.47%, highest precision of 99.02%, highest recall of 90.48%, and highest F1 score of 93.95%. The results of the confusion matrix confirmed that the majority of records classified as Low Risk, Medium Risk, and High Risk were correctly classified, with two High Risk records being classified as Medium Risk. The entire system was also evaluated using the Streamlit app. The tests showed that data validation, risk prediction, explanation of the XAI, generation of recommendations, and dashboard presentation were all parts of the same process. The results complement the primary research goal of creating an explainable human-related cybersecurity risk prediction system. The results, however, are weak due to the small, imbalanced data set, particularly for the High Risk class. In summary, the chapter demonstrates the good performance of CyberRiskXAI as a research prototype. The concluding chapter is the overall conclusion, research contributions, limitations and suggested future improvements.

Chapter 5

Standards, Constraints and Milestones

5.1 Standards and Compliance

CyberRiskXAI is based on best practice set by ISO, IEEE, and NIST. Information security, data protection, access control and secure handling of user information are related to ISO/IEC 27001 and ISO/IEC 27002. The system is based on these principles; it collects only the essential data from the surveys, verifies the inputs prior to processing and it is designed to separate the model development from the web application. If users were to be used in a wide organizational environment, more robust controls like user authentication, role-based access, encrypted communications, and secure data storage would also be necessary.

ISO/IEC 42001 and the NIST AI Risk Management Framework support responsible practices with AI. These frameworks emphasize transparency, accountability, fairness, and

effective risk management. CyberRiskXAI embraces these concepts by comparing a number of machine learning models, examining accuracy, precision, recall, F1 score, and class level results. To explain why the system makes a certain risk prediction, also SHAP and LIME are used. This helps to make the results more understandable and avoids using the model as a "black box" for decision making.

The NIST Cybersecurity Framework is applicable due to its ability to detect unsafe cybersecurity behaviours and give appropriate recommendations. IEEE software engineering and testing practices are reflected by structured model development, train-test splitting, input validation, scenario testing and end-to-end system testing. CyberRiskXAI is an academic prototype and not formally certified or fully compliant. Independent security testing, formal documentation, ongoing monitoring, enhanced privacy measures, and regular system audits would be required for full compliance.

5.2 Sustainability Considerations

The CyberRiskXAI framework was developed to leverage on limited computational resources. It does not involve deep learning because of small size of the dataset, structured dataset and survey based dataset. For this research, there is no clear benefit of using deep learning, as it would need more data, more computing power and more energy. The chosen Logistic Regression method is also simple. It is readily trainable and deployable on standard cloud computing resources. The model was developed using Google Colab, so there is no need for a high-end local computer, GPU or TPU.

The model is trained and then loaded into the Streamlit application. It is not retrained each time a user submits a survey. This saves time, memory and re-use of energy. Input validation, data transformation, prediction, explanation and generation of recommendations constitute a normal use of the application. Modular design also enables a separate update of the model without rebuilding the entire system.

The framework can support increased use as Streamlit enables individual records and uploaded csv files to be processed via web browser. Multiple records can be submitted by enterprise users with an overall risk summary. But the present research prototype is intended just for a small number of users. If you are looking to deploy at a large scale, you might need more cloud hosting capacity, a database, load management and regular performance monitoring. These additions would result in cost and energy consumption. In summary, CyberRiskXAI is quite efficient and environmentally friendly for the current

research field. The simplicity of its model, the reusability of trained file and the web-based structure are also beneficial for long-term maintenance and future growth.

5.3 Societal Impact

There are potential benefits for individuals and organisations to the CyberRiskXAI framework. It can assist users in recognizing the impact of their everyday behaviors like using the same password, clicking on unsafe links, using public Wi-Fi, and having poor privacy habits on their cybersecurity risk. The system does not just issue a risk indicator; it also illustrates the key reasons for the forecast and gives appropriate suggestions. This can help to raise awareness of and promote safer cybersecurity practices. The enterprise-level results can also be used by universities, companies, banks, hospitals and government offices to identify the common weaknesses and plan awareness or training programs.

But if not controlled, the system can have negative effects. A wrong prediction may lead to the user being mis-labelled as High Risk or to a user who is Low Risk becoming complacent. If an organization uses risk labels to assess an employee or student, this can also lead to embarrassment, discrimination, and/or unnecessary monitoring. Personal behavioural data is included in the survey answers. There may thus be privacy concerns due to poor data handling. The web-based system may also fail to benefit individuals with limited access to the Internet or who have low digital skills.

Therefore, it is recommended to use CyberRiskXAI as a supportive tool and not to rely on it as the sole source for taking critical decisions. Results should be subject to human judgment and individuals should be aware of the use of their data and predictions. SHAP and LIME help alleviate some of these risks by improving the interpretability of the predictions. Organizational use also should include access control, informed consent, secure handling of data, and fair use policies. The system has a potential for creating safer behaviour online and for better cybersecurity planning as a whole, but its social impact is reliant on prediction accuracy, responsible use and user protection.

5.4 Ethical Considerations

There are multiple ethical implications in the CyberRiskXAI framework that must be taken into account, such as fairness, privacy and responsible use. Data is collected using self-reported information from the university students in Bangladesh. Thus, the findings do not necessarily reflect the views of people of other ages, professions or countries. There are

also more Medium Risk records than Low Risk and High Risk records. This imbalance can impact the fairness of the model's predictions for each risk group. Stratified data splitting, macro-level performance measures, confusion matrices, and class-level evaluation were used to mitigate this. However, bias can only be eradicated with an increased and more diverse dataset.

Data privacy is another reason as the survey includes details about users' cybersecurity habits. The system only stores the information it requires for the risk assessment and works exclusively in a structured way. Personal or organizational results should never be given to unauthorized users. The use of informed consent, limited access to data, secure storage, and removal of unnecessary personal identifiers are required for wider use. Risk labels should not be used as a punishment, rejection or discrimination tool for students or employees in the system.

Overall, the framework is good for responsible cybersecurity assessment, but more privacy controls, fairness testing, and independent review would be needed for larger scale organizational deployment.

5.5 Security and Privacy Considerations

The CyberRiskXAI framework consumes survey information on user's cybersecurity behaviors. This poses a privacy risk as the answers could identify poor password hygiene, unsafe online habits or other personal security vulnerabilities. Threats include unauthorized access to uploaded CSV files, exposure of individual/organizational reports, malicious input, alterations of the saved model, and attacks against the Streamlit application. There can also be an increased level of risk with cloud deployment when data, access, or software dependencies are not properly managed.

The system at the prototype level implements a number of basic secure design practices. Gathers only what it requires from the survey to assess risk. Checks for missing, incorrect, and inconsistent inputs before processing. The development of the models is isolated from the Streamlit application. Users do not have the option to retrain the model or directly modify it, but rather the website loads a model that has already been trained. This separation minimises unnecessary access to the model-development process. Also, the system requires handling structured data and safeguarding personal and organizational data.

However, these measures aren't sufficient for large-scale deployment. User authentication, role-based access control, encrypted communication, secure database storage, limited

access to reports, activity logs, backup plans and frequent software updates should be part of an enterprise version. The uploaded files should also check the validity of file format and the size of the file before processing. Wherever possible, sensitive identifiers should be removed or anonymized. Universities, companies, banks, hospitals, or government organizations would also have to undergo security testing, vulnerability scanning and penetration testing prior to use. Thus, as a research prototype, CyberRiskXAI offers fundamental security and privacy management, with robust technical and administrative controls needed for enterprise deployment.

5.6 Design Constraints and Component Constraints

CyberRiskXAI is a modular framework, where the model development is performed in Google Colab and the trained model is then plugged into the Streamlit application. This design facilitates the development, testing, and maintenance of the system. But it also brings several constraints. The web application is not capable of automatically retraining or updating the model. If any change in the data set/survey questions/answer mapping/feature order is made, the model needs to be retrained in Google Colab and then changed in the application. The system only concentrates on the human cybersecurity risk derived from surveys. It doesn't watch networks, find live attacks, scan devices, or replace tech security tools.

The present architecture was designed to be a research prototype. While it supports individual input of survey data and enterprise-level processing of csv data, it is not suitable for a very large number of users simultaneously. There are also limited authentication, access control, secure storage and database support in the prototype. Although they would add to the security and scalability of the system, they would also make the system more complex, increase development time, and maintenance requirements.

There are also limitations in the tools and components. Google Colab requires an internet connection and may have session, memory and runtime restrictions. Streamlit is good for a simple web app, but might not work well when handling large files or multiple requests simultaneously. As simple and easy as CSV files are to use, they offer no strong access control, transaction management, or secure multi-user storage. Although SHAP and LIME enhance understanding of a model, it can be more time-consuming to create explanations when a large set of records are being processed simultaneously. It also relies on compatible versions of Python, Scikit-learn, Streamlit, SHAP, and LIME. Modifications to these

libraries could cause model loading or system operation to be impacted.

The design decisions maintained simplicity, low cost, and academic research appropriateness of the system. For large-scale enterprise deployments, however, better infrastructure, database support, automated model management, security controls and optimized processing would be necessary.

5.7 Data Constraints and Computational Constraints

The CyberRiskXAI framework has several data-related limitations. The data set is small and primarily composed of the opinions of university students in Bangladesh as a result of surveys. So the results don't necessarily reflect employees, older users or individuals from other countries. Data are self-reported, and some of the participants may give incomplete, inaccurate or socially acceptable responses rather than the description of their behaviour. Also, there are far more Medium Risk records compared to the number of Low Risk and High Risk records. This imbalance could lead the model to be more indicative of the majority class. This problem was reduced and studied using stratified splitting, macro-level metrics, class-level reports and confusion matrices; however, it cannot be completely removed.

The amount of information that can be gathered is also constrained by privacy restrictions. Results of the cybersecurity survey could show poor security practices, therefore do not include any unnecessary personal identifiers. This would benefit the users but at the same time restrict the access of other personal/organisational features that could enhance prediction. To validate that the model is equally effective in different environments, a larger and more balanced and diverse set of data is required. Current system requires low computing power as it is structured survey data and supervised machine learning and not deep learning. Standard resources provided by Google Colab can be used to train Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression and Random Forest. No GPU/TPU is required. Google Colab, however, has a limit on session duration, memory and storage as well as availability. Furthermore, if multiple users or large CSV files are being processed simultaneously, Streamlit can slow down. For enterprise-level analysis, SHAP and LIME explanations might make processing more time-consuming. While the current prototype is not significantly impacted by these limits, they would be for a large scale deployment, and include enhanced cloud infrastructure, optimized batch processing and more memory.

5.8 Budget Constraints

The CyberRiskXAI framework is designed using a small academic budget. The tools used in the project are mostly free or open source such as Python, Google Colab, Pandas, NumPy, Scikit-learn, SHAP, LIME, Streamlit etc. The survey data were saved in a CSV format, therefore no paid database or special data platform was necessary. A typical computer, Internet connection and web browser was all that was needed for development and testing. This helped to keep the expense of the research prototype down.

But the available budget meant that they could not utilize dedicated servers, paid cloud resources, commercial security tools and professional security testing. Also, a larger study would necessitate additional resources for data collection, recruitment of participants, data storage and maintenance of the system. Additional costs associated with secure hosting, database, authentication, backups, monitoring, technical support and regular security audits would occur with full enterprise deployment. So, this current system is financially appropriate for academic research and small-scale testing, a larger budget would be needed for secure and reliable organizational use.

5.9 Task Breakdown and Project Timeline

The project was completed within the expected thirty-two weeks. Every stage had its own objectives, which served as a foundation for future work. Such a sequence of steps made sure that there would always be continuity and that the research would proceed on time. The first step was requirements analysis and system design. It included the careful review of existing literature to reveal gaps in previous research and formulate system requirements. During this period, the scope of research, objectives and data collection strategy were finalized. The second phase involved the design of the cyber security awareness questionnaire and collection of response from the students of Bangladeshi universities. The collected dataset was then analyzed for completeness and processed further. The next stage included data preprocessing and feature engineering. The survey responses were cleaned and recoded to numeric values and converted to cybersecurity risk scores. The processed data set was then ready to undergo machine learning analysis. The fourth phase was development and evaluation of Logistic Regression model. The model was trained on the processed dataset and evaluated by accuracy, precision, recall and F1-score. The system integration was done for the best configuration.

The fifth phase concerned the implementation of the CyberRiskXAI web-application. Stepwise integration of modules for data upload, preprocessing, prediction, SHAP and LIME explanations and recommendation was done. The system was tested several times to improve the functionality and reliability. The final stage included system validation, results analysis, documentation and thesis preparation. The experimental results were carefully interpreted before the final report and presentation materials were completed.

Phase	Main Activities	Duration
Phase 1	Project planning and requirement analysis	Weeks 1–4
Phase 2	Literature review and research gap analysis	Weeks 5–8
Phase 3	Survey design and data collection	Weeks 9–12
Phase 4	Data preprocessing and feature preparation	Weeks 13–16
Phase 5	Model development and training	Weeks 17–20
Phase 6	Model evaluation and refinement	Weeks 21–23
Phase 7	System implementation and XAI integration	Weeks 24–27
Phase 8	Testing, validation, and improvements	Weeks 28–30
Phase 9	Documentation, thesis writing, and presentation	Weeks 31–32

Table 2: Task Breakdown

5.10 Milestones and Deliverables

Many key milestones were reached during the thirty-two weeks' project. Each milestone indicated the end of each phase of the development process. These deliverables facilitated the delivery of the intended results towards the research goals.

The first milestone was the successful completion of planning and requirements analysis of the project. This phase defined the research scope, objectives, system requirements and overall development strategy. The main deliverables were the software requirement specification and the project proposal. The second milestone was the literature review and the discovery of the research gap. Existing approaches and limitations were analyzed based on previous studies. The deliverable for this phase was the final literature review chapter. The third milestone consisted of the design of the cybersecurity awareness questionnaire and the collection of the survey responses. The data were collected from the university students of Bangladesh by structured questionnaire. The deliverable was a finished survey database for further analysis. The fourth milestone was to pre-process the data that was collected and prepare features to be used in machine learning algorithms. Survey responses

were cleaned and encoded into cybersecurity risk score. The last machine learning ready dataset was obtained during this phase. The fifth milestone was the development and evaluation of the prediction model of Logistic Regression. The model was trained, evaluated and tested using several metrics for performance. Deliverables included the trained model, evaluation results and performance reports. The sixth milestone was the launch of the CyberRiskXAI web application. Different modules were integrated for uploading data, pre-processing, prediction of cybersecurity risk, SHAP and LIME explanations, and personalized recommendations. The end result was a fully functional prediction system, running on the web.

The last milestone was system testing, documentation and thesis writing. All experimental results were analyzed and the final report was prepared for submission. The deliverables were the final thesis, project documentation and final presentation.

Milestone	Weeks	Deliverable
Project planning completed	1–4	Project proposal and Software Requirement Specification (SRS)
Literature review completed	5–8	Chapter 2: Literature Review
Survey and data collection completed	9–12	Cybersecurity awareness survey dataset
Data preprocessing completed	13–16	Machine learning-ready dataset
Model development completed	17–20	Trained Logistic Regression model
Model evaluation completed	21–23	Accuracy, Precision, Recall, F1-score, and evaluation report
System implementation completed	24–27	CyberRiskXAI web application with SHAP and LIME integration
Testing and validation completed	28–30	Testing report and validated system
Thesis and presentation completed	31–32	Final thesis, project documentation, and presentation slides

Table 3: Milestone and Deliverables

5.11 Gantt Chart

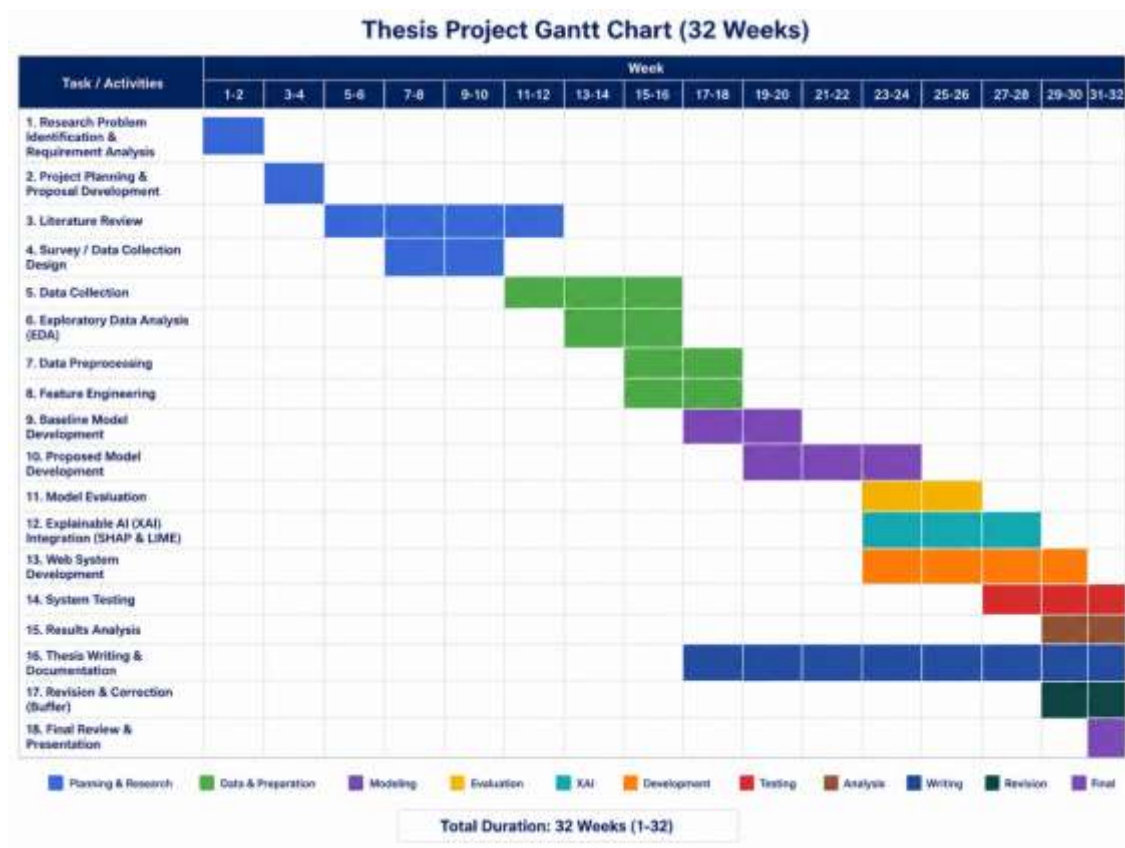


Figure 10: Gantt Chart

5.12 Summary

This chapter discussed the main standards, ethical issues, security concerns, and limitations of the CyberRiskXAI framework. The key constraints include a small and imbalanced dataset, limited computing resources, basic security controls, and limited budget. These issues were reduced by using stratified splitting, lightweight machine learning models, free open-source tools, input validation, and a modular system design. Although the system performs well as a research prototype, stronger infrastructure, more diverse data, and improved security would be needed for large-scale use.

Chapter 6

Conclusion

6.1 Summary of the Study

This study proposes the framework of CyberRiskXAI; which is a machine learning and Explainable AI based framework for predicting the cybersecurity risk among the university students in Bangladesh. The research was based on the dangers arising out of everyday user action. Some of the behaviors were: use of the same password for multiple sites, weak 2FA, unsafe links, public Wi-Fi, not being careful about privacy, and installation of untrusted software. The primary objective was the classification of users into Low Risk, Medium Risk or High Risk and the rationale for each classification. The survey data was gathered, cleaned, encoded and converted into numbers. Cyber Safety Scores and risk labels were then created. The data were split into 80% training data and 20% testing data. The machine learning models tested were: Balanced Decision Tree, Gaussian Naive Bayes, Logistic Regression and Random Forest. Logistic Regression got the best results with 97.47% accuracy and 93.95% F1 score. The model predictions were explained using SHAP and LIME. A Streamlit web application was created for the chosen model. The final system will be able to evaluate each user, process organizational data, present results of risk, explain key factors and make cyber security recommendations. In general, the primary objective of the study was realized and an understandable and practically applicable system for human-related cybersecurity risk assessment was developed.

6.2 Key Contributions

This research contributed in a number of important ways. First, it created a full cybersecurity risk prediction workflow out of survey data. The workflow includes data cleaning, answer encoding, score creation, risk labeling, model training and testing. Secondly, a comparison of four machine learning models was made on the same set of data. Logistic Regression was chosen as the best model as it had the highest overall performance. Thirdly, to explain the predictions, SHAP and LIME were added. This helped to interpret the results. Fourthly, a Streamlit web application was created to connect the model. The system evaluates users on an individual basis and can evaluate several records for enterprise level evaluation. Finally, the system offers suggestions based on the weaknesses in the cybersecurity practices in the survey responses. Such contributions brought forth a

framework for the study of human related cybersecurity risk among the university students in Bangladesh that is practical and explainable.

6.3 Limitations of the Study

There are some limitations associated with this study. The dataset was relatively small and was mainly collected from university students in Bangladesh. Therefore, the findings may not represent other age groups, professions, or countries. The dataset also contained more Medium Risk records than Low Risk and High Risk records. This class imbalance may reduce the model's performance for the smaller groups. The survey data were self-reported, so some answers may not fully reflect the users' actual behaviour. Another limitation is that the scoring and risk-labeling process was created manually using fixed values for each survey answer. This may include researcher judgement and may not always reflect the exact importance of every cybersecurity behaviour. In future versions, the scoring process should be automated and updated using validated expert weights or data-driven methods. Only four machine learning models were used in the research. The system was tested as a research prototype rather than in a large real-world organization. The Streamlit application also has limited security, storage, and user-management features. SHAP and LIME improved the explanation of predictions, but some non-technical users may still find the explanations difficult to understand. Therefore, the findings are useful for the current research, but more data, automatic scoring, stronger system features, and wider testing are needed before large-scale deployment.

6.4 Future Work

Future research should enhance CyberRiskXAI by expanding the dataset, which should be larger and more balanced. Sample should be taken from various universities, professions, age groups, and regions. This would assist the system to be more accurate in making predictions particularly for users in the High Risk category.

It is also possible to integrate the system with AI services like ChatGPT or Gemini using an API. The prediction result and XAI result generated by the machine learning could be delivered to the AI module. It would then be able to give real-time and individual cyber security recommendations according to the user's weak security practices. It can also be secured with a security bot. The bot might scan the web application, software requirements, input forms, and system configuration for prevalent vulnerabilities. It may uncover

problems like inadequate authentication, outdated libraries, unsafe file uploads, inadequate input validation or potentially flawed access controls. The bot might then alert and recommend appropriate remedies. This should only be performed on vetted systems and be oriented toward defensive security testing. A secure login, role-based access, encrypted storage, database, and activity logs are additional features that can enhance the web application. Future versions may have the ability to provide department-level reports, risk trend reports, regular model updates, and explanations in a non-technical way. Finally, the complete framework should be tested in real Universities and organizations before its large-scale use.

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Appendix

These are the following questions for the conducted survey:

Section A: Demographic Information

1. Gender
2. Age
3. Department
4. Year of Study

Section B: Cybersecurity Awareness and Behaviour

5. Do you reuse passwords across multiple accounts?
6. Do you use Two-Factor Authentication (2FA) for your accounts?
7. Do you click on links from emails or messages from unknown sources?
8. How do you rate your awareness of phishing and online scams?
9. Do you share your passwords with friends or family?
10. Do you use public Wi-Fi for sensitive activities (such as online banking)?
11. Do you use antivirus software on your devices?
12. Do you use unique passwords for important accounts?
13. Do you read privacy policies before installing applications?
14. Do you verify website security (HTTPS) before entering sensitive information?
15. Do you educate yourself about online threats (e.g., reading articles or watching tutorials)?
16. Do you use a password manager?
17. Do you change passwords periodically (every 3–6 months)?
18. Do you verify the authenticity of software before installing it?
19. Do you use separate devices for sensitive activities (such as banking or exams)?
20. What do you think about your cybersecurity awareness level?